

2-Day Moving Average — Complete Line-by-Line Calculation

Only the Moving Average column is considered. The last/error column is NOT used.

1. What is a Moving Average?

A moving average takes a fixed number of nearby sales values, adds them, and divides by the number of values. Here we use a **2-Day Moving Average**, so we use 2 sales values at a time.

Formula:

$$\text{Moving Average} = (\text{First Sales} + \text{Second Sales}) \div 2$$

For this example, Day 1 uses Day 1 and Day 2. Day 2 uses Day 2 and Day 3, and so on.

2. Day-by-Day Calculation

Day 1

$$\text{Sales on Day 1} = 120$$

$$\text{Sales on Day 2} = 125$$

$$\begin{aligned}\text{Moving Average} &= (120 + 125) \div 2 \\ &= 245 \div 2 \\ &= \mathbf{122.5}\end{aligned}$$

Day 2

$$\text{Sales on Day 2} = 125$$

$$\text{Sales on Day 3} = 123$$

$$\begin{aligned}\text{Moving Average} &= (125 + 123) \div 2 \\ &= 248 \div 2 \\ &= \mathbf{124}\end{aligned}$$

Day 3

$$\text{Sales on Day 3} = 123$$

$$\text{Sales on Day 4} = 130$$

$$\begin{aligned}\text{Moving Average} &= (123 + 130) \div 2 \\ &= 253 \div 2 \\ &= \mathbf{126.5}\end{aligned}$$

Day 4

$$\text{Sales on Day 4} = 130$$

$$\text{Sales on Day 5} = 128$$

$$\begin{aligned}\text{Moving Average} &= (130 + 128) \div 2 \\ &= 258 \div 2 \\ &= \mathbf{129}\end{aligned}$$

Day 5

$$\text{Sales on Day 5} = 128$$

$$\text{Sales on Day 6} = 135$$

$$\begin{aligned}\text{Moving Average} &= (128 + 135) \div 2 \\ &= 263 \div 2 \\ &= \mathbf{131.5}\end{aligned}$$

Day 6

$$\text{Sales on Day 6} = 135$$

$$\text{Sales on Day 7} = 140$$

$$\begin{aligned}\text{Moving Average} &= (135 + 140) \div 2 \\ &= 275 \div 2 \\ &= \mathbf{137.5}\end{aligned}$$

2. Day-by-Day Calculation — Continued

Day 7

Sales on Day 7 = 140

Sales on Day 8 = 138

Moving Average = $(140 + 138) \div 2$

= $278 \div 2$

= **139**

Day 8

Sales on Day 8 = 138

Sales on Day 9 = 145

Moving Average = $(138 + 145) \div 2$

= $283 \div 2$

= **141.5**

Day 9

Sales on Day 9 = 145

Sales on Day 10 = 150

Moving Average = $(145 + 150) \div 2$

= $295 \div 2$

= **147.5**

Day 10

Sales on Day 10 = 150

Sales on Day 11 = 148

Moving Average = $(150 + 148) \div 2$

= $298 \div 2$

= **149**

Day 11

Sales on Day 11 = 148

Sales on Day 12 = 155

Moving Average = $(148 + 155) \div 2$

= $303 \div 2$

= **151.5**

Day 12

Sales on Day 12 = 155

Sales on Day 13 = 160

Moving Average = $(155 + 160) \div 2$

= $315 \div 2$

= **157.5**

2. Day-by-Day Calculation — Continued

Day 13

Sales on Day 13 = 160

Sales on Day 14 = 158

Moving Average = $(160 + 158) \div 2$

= $318 \div 2$

= **159**

Day 14

Sales on Day 14 = 158

Sales on Day 15 = 165

Moving Average = $(158 + 165) \div 2$

= $323 \div 2$

= **161.5**

Day 15

Sales on Day 15 = 165

Sales on Day 16 = 170

Moving Average = $(165 + 170) \div 2$

= $335 \div 2$

= **167.5**

Day 16

Sales on Day 16 = 170

Sales on Day 17 = 168

Moving Average = $(170 + 168) \div 2$

= $338 \div 2$

= **169**

Day 17

Sales on Day 17 = 168

Sales on Day 18 = 175

Moving Average = $(168 + 175) \div 2$

= $343 \div 2$

= **171.5**

Day 18

Sales on Day 18 = 175

Sales on Day 19 = 180

Moving Average = $(175 + 180) \div 2$

= $355 \div 2$

= **177.5**

2. Day-by-Day Calculation — Continued

Day 19

Sales on Day 19 = 180

Sales on Day 20 = 178

Moving Average = $(180 + 178) \div 2$

= $358 \div 2$

= **179**

Day 20

Sales on Day 20 = 178

To calculate the 2-day moving average, we would need Day 21 sales.

Moving Average = $(178 + \text{Day 21 Sales}) \div 2$

Day 21 sales are not provided.

Therefore, Day 20 Moving Average = **Blank / Cannot be calculated.**

3. Complete Moving Average Table

Day	Sales	Calculation	Moving Average
1	120	$(120 + 125) \div 2$	122.5
2	125	$(125 + 123) \div 2$	124
3	123	$(123 + 130) \div 2$	126.5
4	130	$(130 + 128) \div 2$	129
5	128	$(128 + 135) \div 2$	131.5
6	135	$(135 + 140) \div 2$	137.5
7	140	$(140 + 138) \div 2$	139
8	138	$(138 + 145) \div 2$	141.5
9	145	$(145 + 150) \div 2$	147.5
10	150	$(150 + 148) \div 2$	149
11	148	$(148 + 155) \div 2$	151.5
12	155	$(155 + 160) \div 2$	157.5
13	160	$(160 + 158) \div 2$	159
14	158	$(158 + 165) \div 2$	161.5
15	165	$(165 + 170) \div 2$	167.5
16	170	$(170 + 168) \div 2$	169
17	168	$(168 + 175) \div 2$	171.5
18	175	$(175 + 180) \div 2$	177.5
19	180	$(180 + 178) \div 2$	179
20	178	Needs Day 21	Blank

4. The Easy Rule to Remember

Take 2 consecutive sales values → Add them → Divide by 2.

Example: Day 8 = $(138 + 145) \div 2 = 283 \div 2 = 141.5$

Important: Do NOT use the last column for this calculation. It is not part of the moving-average calculation.

5. Why Day 20 is Blank

A 2-day moving average for Day 20 would require Day 20 and Day 21. Since Day 21 is missing, the calculation cannot be completed.